

Lengths and Ratios

Diana Tolu

June, 2025

Problem 1. (G3, IMO Shl 2023)

Let $ABCD$ be a cyclic quadrilateral with $\angle BAD < \angle ADC$. Let M be the midpoint of the arc CD not containing A . Suppose there is a point P inside $ABCD$ such that $\angle ADB = \angle CPD$ and $\angle ADP = \angle PCB$.

Prove that lines AD , PM , and BC are concurrent.

Problem 2. (Problem 5, IMO 1999) Two circles ω_1 and ω_2 are contained inside the circle ω , and are tangent to G at the distinct points M and N , respectively. ω_1 passes through the center of ω_2 . The line passing through the two points of intersection of ω_1 and ω_2 meets ω at A and B . The lines MA and MB meet ω_1 at C and D , respectively.

Prove that CD is tangent to ω_2 .

Problem 3. (Problem 2, IMO 1982)

A non-isosceles triangle $A_1A_2A_3$ has sides a_1, a_2, a_3 with the side a_i lying opposite to the vertex A_i . Let M_i be the midpoint of the side a_i , and let T_i be the point where the inscribed circle of triangle $A_1A_2A_3$ touches the side a_i . Denote by S_i the reflection of the point T_i in the interior angle bisector of the angle A_i . Prove that the lines M_1S_1, M_2S_2 and M_3S_3 are concurrent.

Problem 4. (G3, IMO Shl 2000)

Let O be the circumcenter and H be the orthocenter of acute triangle ABC . Show there exist points D, E, F on sides BC, CA , and AB respectively such that:

$$OD + DH = OE + EH = OF + FH$$

and the lines AD, BE, CF are concurrent.

Problem 5. (APMO 2000) Let ABC be a triangle. Let M and N be the points in which the median and the angle bisector, respectively, at A meet the side BC . Let Q and P be the points in which the perpendicular at N to NA meets MA and BA , respectively, and O the point in which the perpendicular at P to BA meets AN produced. Prove that QO is perpendicular to BC .

Problem 6. (Taiwan TST) Given a triangle $\triangle ABC$ with orthocenter H . On its circumcenter, choose an arbitrary point P (other than A, B, C) and let M be the mid-point of HP . Now, we find three points D, E, F on the line BC, CA, AB , respectively, such that $AP \parallel HD, BP \parallel HE, CP \parallel HF$. Show that D, E, F, M are collinear.

Problem 7. (USA TSTST) Acute triangle ABC is inscribed in circle ω . Let H and O denote its orthocenter and circumcenter, respectively. Let M and N be the midpoints of sides (AB) and (BC) , respectively. Rays MH and NH meet ω at P and Q , respectively. Lines MN and PQ meet at R . Prove that $OA \perp RA$.

Problem 8. (Iranian TST) Let ABC be a triangle ($\angle A \neq 90^\circ$). BE, CF are the altitudes of the triangle. The bisector of $\angle A$ intersects EF, BC at M, N . Let P be a point such that $MP \perp EF$ and $NP \perp BC$. Prove that AP passes through the midpoint of BC .

Problem 9. (2, RMM 2019) Let $ABCD$ be an isosceles trapezoid with $AB \parallel CD$. Let E be the midpoint of AC . Denote by ω and Ω the circumcircles of the triangles ABE and CDE , respectively. Let P be the crossing point of the tangent to ω at A with the tangent to Ω at D . Prove that PE is tangent to Ω .

Problem 10. Let ABC be a triangle. Let D be an arbitrary point on BC (D different from B and C). Let K, L be the points on BC such that AD is the common internal angle bisector of angles $\angle BAK$ and $\angle CAL$. Let M, N be the circumcenters of triangles ABD and ACD , respectively. Prove that the intersection of KN and LM lies on the circumcircle of triangle ABC .

Problem 11. (G5, IMO Shl 2018) Let ABC be a triangle with circumcircle Ω and incentre I . A line ℓ intersects the lines AI, BI, CI at points D, E, F , respectively. The perpendicular bisectors of $(AD), (BE), (CF)$ determine a triangle Δ . Show that the circumcircle of the triangle Δ is tangent to Ω .

Problem 12. (USAMO 2015) Quadrilateral $APBQ$ is inscribed in circle ω with $\angle P = \angle Q = 90^\circ$ and $AP = AQ < BP$. Let X be a variable point on segment PQ . Line AX meets ω again at S (other than A). Point T lies on arc AQB of ω such that $\overline{XT}$ is perpendicular to $\overline{AX}$. Let M denote the midpoint of chord $\overline{ST}$. As X varies on segment PQ , show that M moves along a circle.

Problem 13. (China TST) Let $ABCD$ be a convex quadrilateral, the incenters of $\triangle ABC$ and $\triangle ADC$ are I, J , respectively. It is known that AC, BD, IJ concurrent at a point P . The line perpendicular to BD through P intersects with the outer angle bisector of $\angle BAD$ and the outer angle bisector $\angle BCD$ at E, F , respectively. Show that $PE = PF$.

Problem 14. (USA TSTST) Let ABC be a triangle with incenter I . Points K and L are chosen on segment BC such that the incircles of $\triangle ABK$ and $\triangle ABL$ are tangent at P , and the incircles of $\triangle ACK$ and $\triangle ACL$ are tangent at Q . Prove that $IP = IQ$.